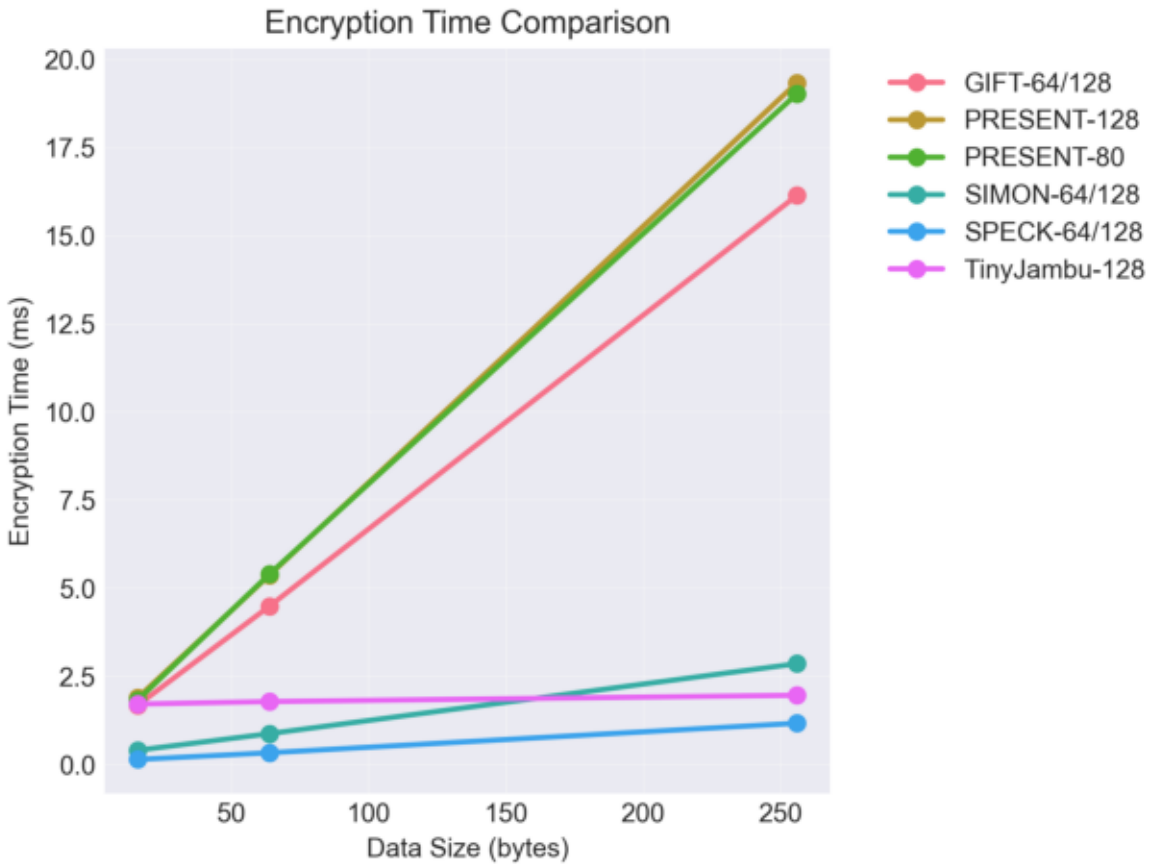
IoT Cryptography Project

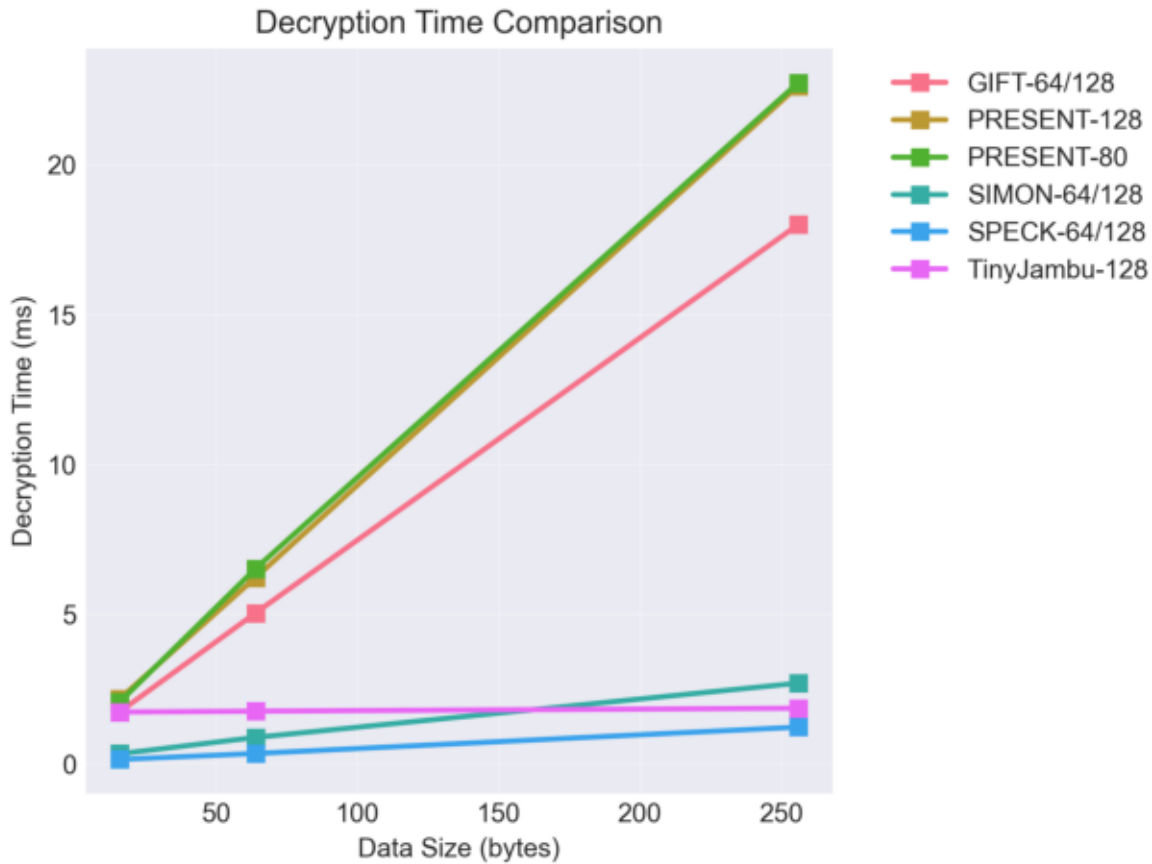
Performance & Security Analysis

Generated: 2026-04-02 13:05:02

Encryption Time

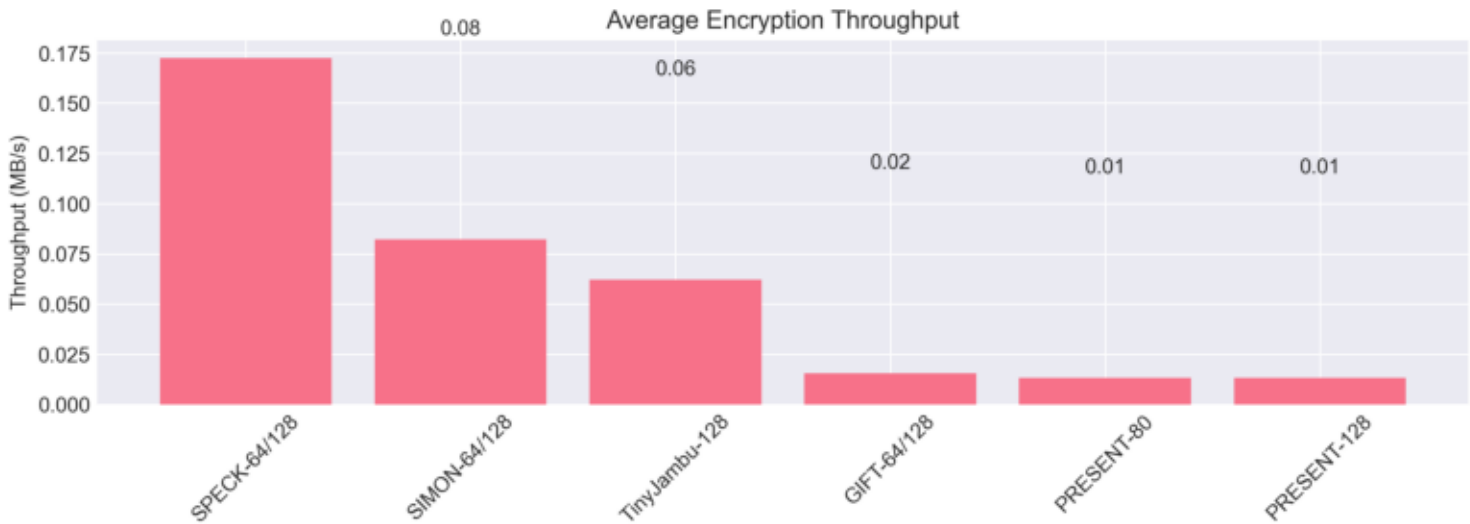


Decryption Time

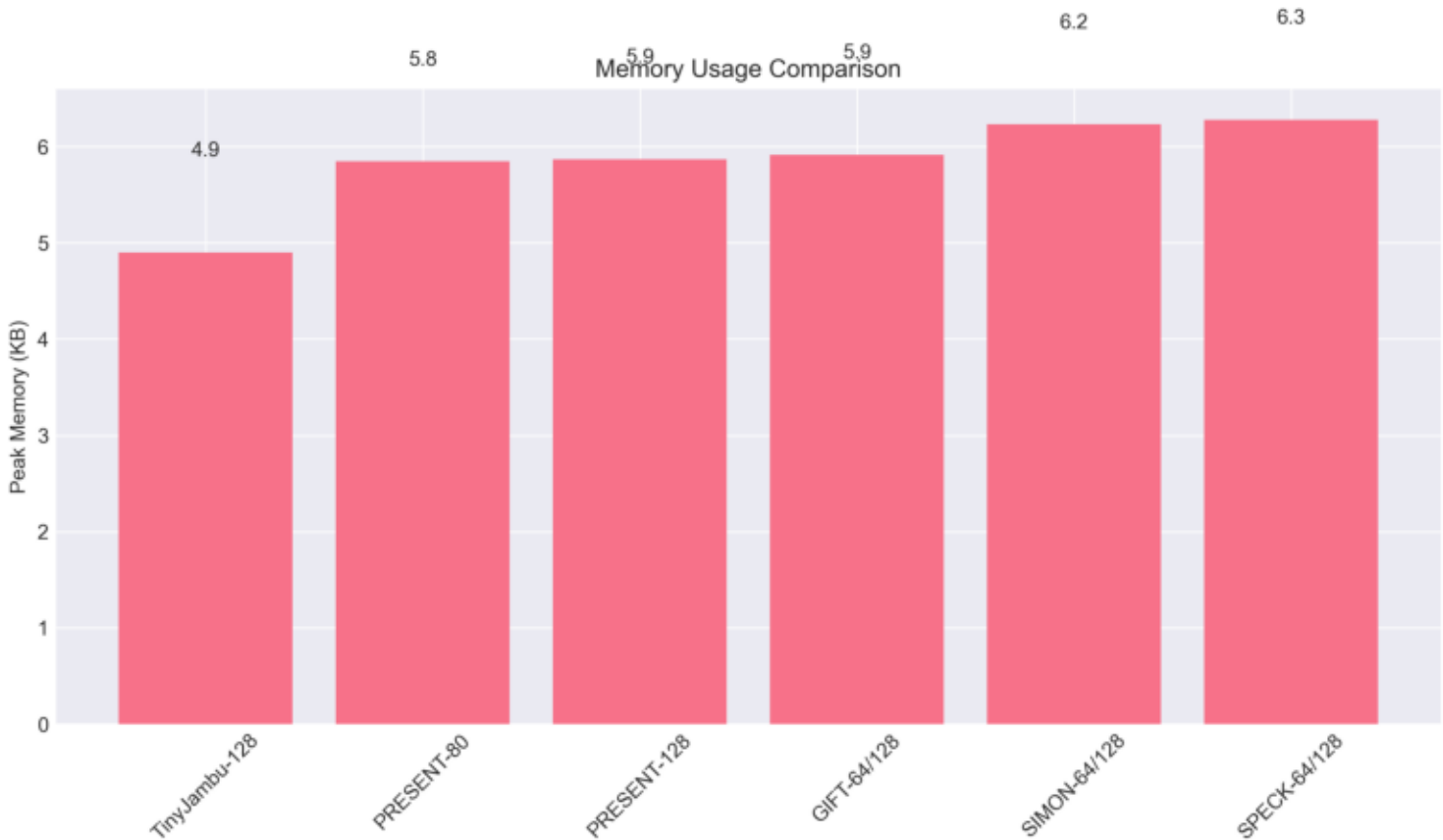


Throughput

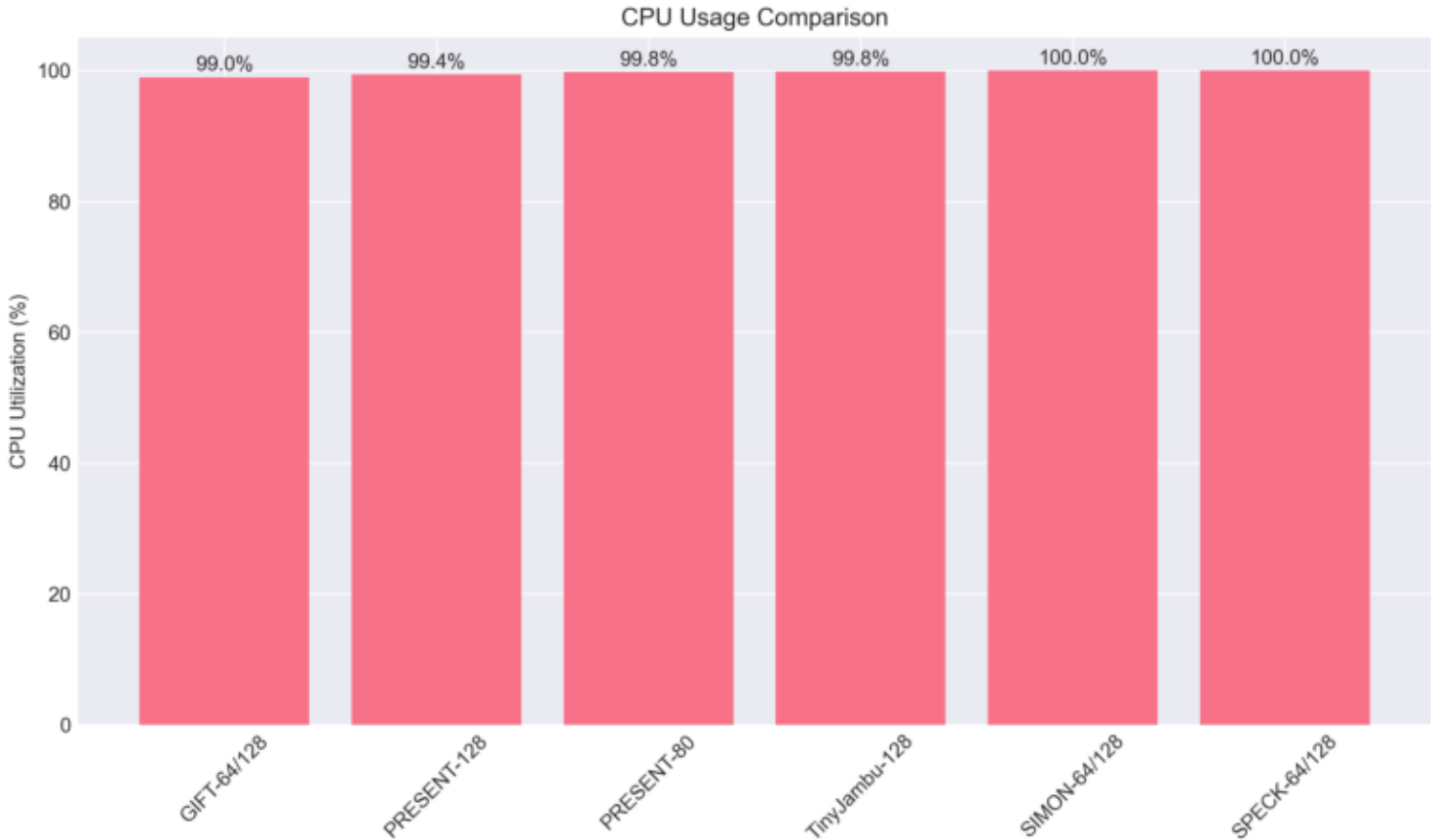
0.17



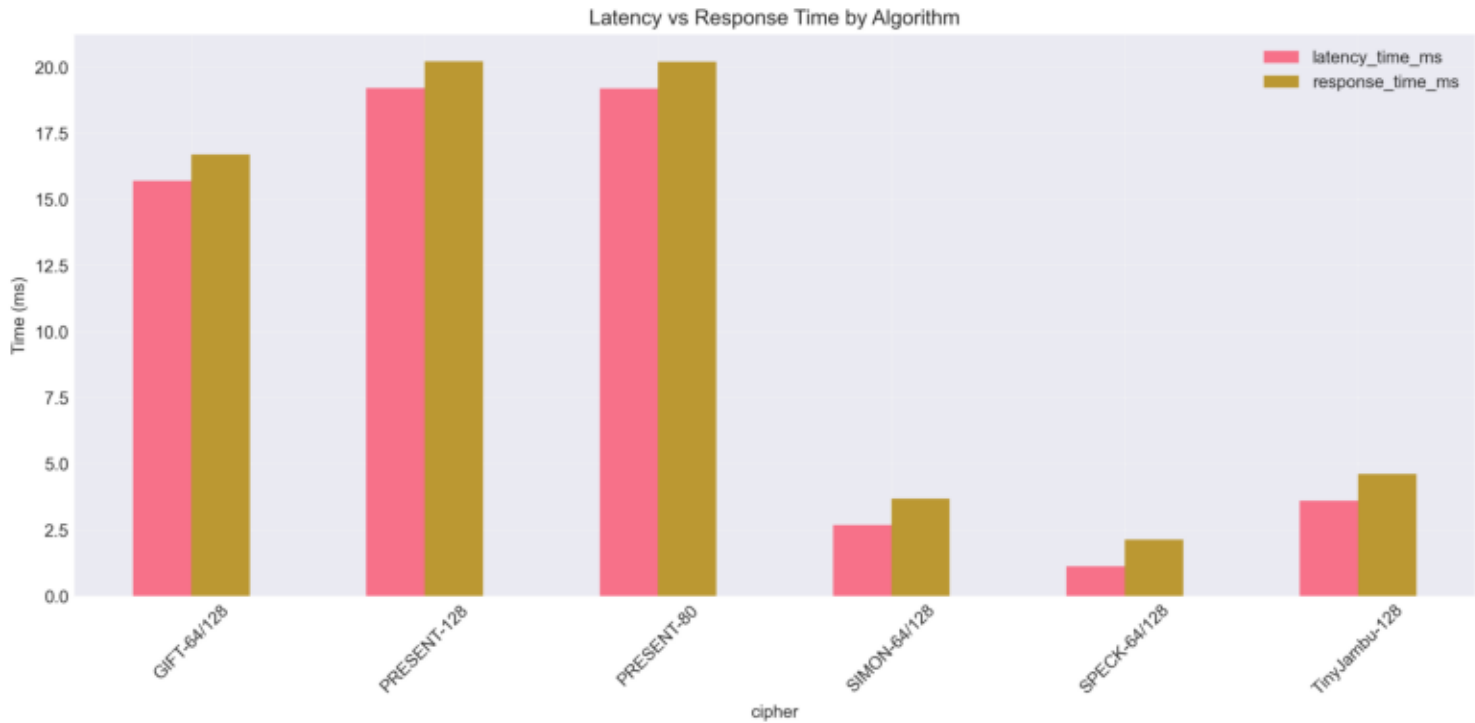
Memory


Memory Usage Comparison
Peak Memory (KB)
0
1
2
3
4
5
6
4.9
5.8
5.9
5.9
6.2
6.3
TinyJambu-128
PRESENT-80
PRESENT-128
GIFT-64/128
SIMON-64/128
SPECK-64/128

Cpu


CPU Usage Comparison
CPU Utilization (%)
0
20
40
60
80
100
99.0%
99.4%
99.8%
99.8%
100.0%
100.0%
GIFT-64/128
PRESENT-128
PRESENT-80
TinyJambu-128
SIMON-64/128
SPECK-64/128

Latency


Latency vs Response Time by Algorithm
latency_time_ms
response_time_ms
Time (ms)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
GIFT-64/128
PRESENT-128
PRESENT-80
SIMON-64/128
SPECK-64/128
Tiny-Jambu-128
cipher

Attack Analysis Results

Attack	Target	Expected	Observed	Blocked	Status
Replay Attack	Packet Replay	Replay should be rejected	Replay attack detected	True	□ Protected
MITM Tampering	Data Integrity	Modified packet should fail MAC	MITM tampering result not available	True	□ Protected
Brute Force (Toy)	Toy 8-bit Key	Key should be recoverable	Recovered: True	False	□□ Demo Only
Brute Force (Real)	128-bit Keys	Should be infeasible	Theoretically infeasible	True	□ Resistant